

Two games, two methods

Hole punching and cube counting are the two you own first. They are procedures, not talent, which is exactly why they are worth the most to you right now. Nobody is naturally good at these. People are practised at them.

HOW TO USE THIS FILE

Open it in GoodNotes or Notability and write straight on the pages with your stylus. Two things make it work properly:

1. DUPLICATE THE PRACTICE PAGES.

Long press the page thumbnail and duplicate it. Fifteen grids becomes thirty, then sixty. You never run out and you never reuse a dirty grid.

2. KEEP THE FINISHED PAGES.

When you miss one, the page shows you exactly where the method broke, which is worth more than the right answer was.

On paper it works the same way. Print the last two pages as many times as you need.

Hole punching: the grid method

Draw the grid every single time. Even when you think you can see the answer, especially then. The whole point is to take the work out of your head and put it on the paper, so that nothing depends on you holding four folds in memory at once.

1. **DRAW** a 4 by 4 grid. Sixteen boxes. Do this before you look at the folds.
2. **HOLE** Mark where the punch goes on the folded paper. A circle, not a dot, so you can still see it once you start reflecting.
3. **FOLD** Find the fold line. It is always along the edge of the paper that moved.
4. **REFLECT** Mirror the hole across that fold line.
5. **CHECK** Work backwards through the folds, one at a time, in reverse order.

THE RULE PEOPLE GET WRONG

Only holes on the moving piece of paper reflect. A hole sitting on the part that never moved stays exactly where it is. If you are consistently one hole over, this is almost always why.

RIGHT TO LEFT

Work the folds right to left, the reverse of the order they were made in. Same direction every time, so it stops being a decision and becomes a reflex.

Cube counting

The T chart

One rhythm, every time, and you never recount. **Bottom to top, front to back, left to right.** Say it out loud while you drill until it is boring.

1. **DRAW** the T chart with rows 0 through 5, one row per number of painted sides.
2. **NUMBER** every cube in the figure, following the rhythm. Every cube gets a number, including the ones you cannot fully see.
3. **TOP** Count the cube sitting on top next, even a floating one.
4. **TALLY** each cube into its row as you go. Finish a row of the figure before you move on.

THE BOTTOM FACE

The face sitting on the table is not painted. That is the single most common miscount on this game, and it costs a whole row of the chart when you get it wrong.

TALLY AS YOU GO

Do not count the whole figure and then fill the chart. Number a cube, decide its painted faces, mark the row, move on. Counting twice is how the recount spiral starts.

Why these two first

Every other game in this section asks you to see something. These two ask you to follow a procedure, which means they are the two you can be certain about on a bad day. Once the grid and the chart are automatic, they are worth 13 to 15 out of 15 and they hold that number under time pressure, which is more than can be said for anything else on the section.

So they come first, and they come daily. Short and daily beats long and occasional, every time. Ten minutes of grids every morning will do more than an hour once a week, because what you are building is a reflex, and reflexes are built by repetition rather than by duration.

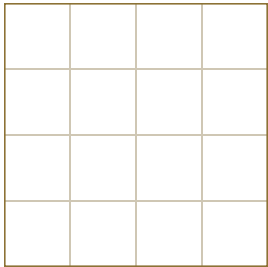
WHEN YOU MISS ONE

Do not just look up the answer. Go back to the page you drew and find the step where it broke. It is almost always the same step twice. That is the thing worth knowing, and it is the reason to keep the pages you write on.

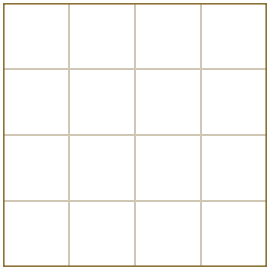
PRACTICE

Hole punching grids

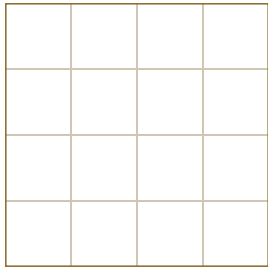
One grid per question. Draw before you decide. Duplicate this page whenever you need more.



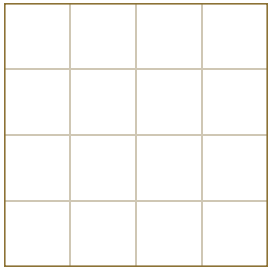
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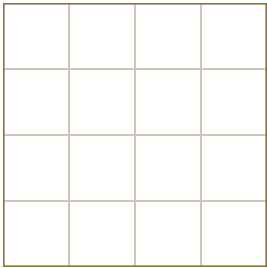
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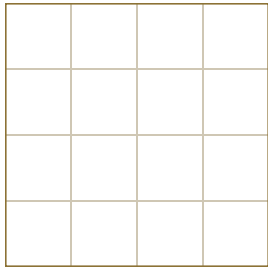
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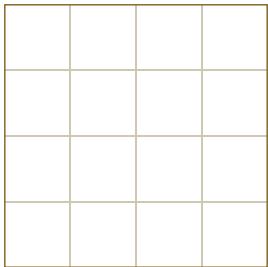
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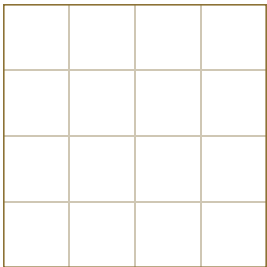
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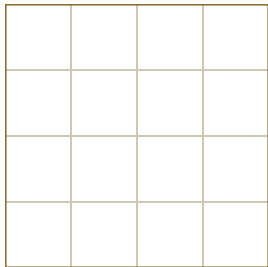
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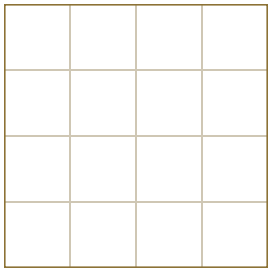
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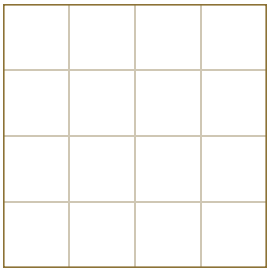
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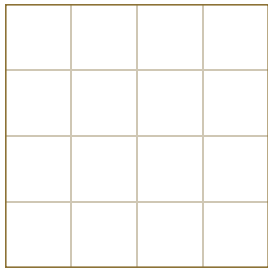
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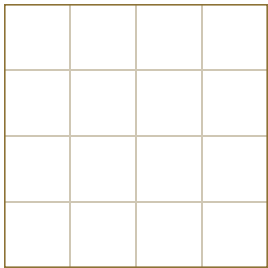
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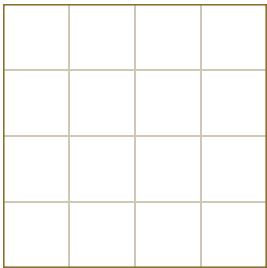
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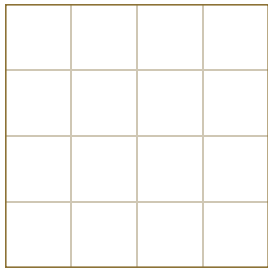
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15

PRACTICE

Cube counting T charts

One chart per figure. Number every cube, then tally as you go. Duplicate this page whenever you need more.

SIDES	CUBES	SIDES	CUBES	SIDES	CUBES	SIDES	CUBES
0		0		0		0	
1		1		1		1	
2		2		2		2	
3		3		3		3	
4		4		4		4	
5		5		5		5	
1		2		3		4	
SIDES	CUBES	SIDES	CUBES	SIDES	CUBES		
0		0		0			
1		1		1			
2		2		2			
3		3		3			
4		4		4			
5		5		5			
5		6		7			

Bottom to top, front to back, left to right. The face on the table is never painted.